



D(4th Sm.)-Computer Science-MDC/MN-2/CCF

2025

COMPUTER SCIENCE — MDC

Paper : MN-2

(Problem Solving using C)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer **any five** questions from the following : 2×5
- (a) What are header files in C? Give an example and briefly state its use.
 - (b) Compare the behaviour of the prefix (++P) and Postfix (P++) increment operators in C.
 - (c) What are some common modes of opening files in C?
 - (d) What is a Pointer in C? Why is it significant in Programming?
 - (e) What is the role of switch statement in controlling program flow?
 - (f) What is the role of the continue statement?
 - (g) What are the differences between the while loop and the do-while loop in C?
 - (h) What is a function prototype?

Section - A

Answer **any three** questions.

- 2. What are different storage classes in C? Explain with example. 5
- 3. Explain difference between call by value and call by address. 5
- 4. What is Union? How it is different from structure? 3+2
- 5. What are command-line arguments in C? Illustrate with an example that adds two numbers passed from the command line. 2+3
- 6. Draw a flowchart for converting a temperature value from Celsius to Fahrenheit. 5
- 7. What is top-down approach of problem solving? How modular programming is related to the same? 3+2

Please Turn Over

(2923)

Section - B

Answer *any five* questions.

8. (a) What are the different phases of compilation in C?
 (b) Write a C program to calculate the sum of first N natural numbers, without using any formula.
 (c) Mention any two differences between compiler and interpreter. 3+5+2
9. (a) Write a program in C to check whether an integer is prime or composite.
 (b) Briefly describe about relational operators in C with examples. 6+4
10. (a) Design an algorithm for checking if an integer is a palindrome.
 (b) Write a C program that reads an integer n and displays the following pattern, input = 5. 5+5

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11. (a) Write a function in C to check whether an element is present in an array or not.
 (b) Briefly explain row-major order and column major order of arranging elements in a 2D-array. 6+4
12. (a) Explain purpose of strcat() and strcmp() with proper examples.
 (b) What do you mean by nested structure? Give an example.
 (c) What is the meaning of the declaration : int*p[10], ? 4+4+2
13. (a) Differentiate between malloc() and calloc() functions. What is the purpose of free() function?
 (b) What is dangling pointer problem?
 (c) Explain usage of void pointer in C. (3+1)+2+4
14. (a) How is append operation implemented for file handling in C?
 (b) What is the purpose of fseek() and ftell()? Explain with examples. 4+(4+2)